# Non-Redundant and Symmetry-Adapted Coordinates for Black-Box Molecular Structure Refinement

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**Abstract**

Semiexperimental equilibrium structures combine high-resolution rotational spectroscopy with computed vibrational and electronic corrections. The least-squares refinement depends on the coordinate model used to connect observables, constraints, and structural parameters. Z-matrices are compact for simple acyclic molecules; in rings, fused systems, and bridged frameworks they introduce reference-tree choices, derived ring quantities, and manually selected symmetry relationships.

We present `SEfit`, a black-box structure-refinement framework in which the input structure is Cartesian and the active coordinate model is generated from molecular topology and symmetry. Two representations are available. The first is a non-redundant symmetry-adapted generalized-internal-coordinate basis built from homogeneous primitive blocks of stretches, bends, torsions, out-of-plane motions, and ring coordinates. The second is a symmetry-adapted Cartesian displacement basis obtained after exact removal of translations and rotations. In both representations all atoms

enter through the same Cartesian topology, and only the totally symmetric active subspace is refined.

The chemical interface remains internal-coordinate based. Constraints, predicates, and parameter classes are specified as primitive relations because bond lengths, valence angles, dihedrals, and local hydrogen frames are the quantities that carry chemical meaning. These relations are projected onto either active basis through the Wilson matrix or the corresponding primitive-coordinate derivative projector, preserving chemically defined constraints and rigorous uncertainty propagation.

`SEfit` integrates moment-of-inertia fitting, vibrational and electronic corrections, predicate observations, parameter classes, reduced-dimensionality hydrogen models, statistical diagnostics, and human-readable reports within a single Cartesian/topology-based refinement workflow.

Glycolaldehyde is used as a standard acyclic reference. Glycine I and II test constrained refinements with, respectively, a $C_s$ symmetry model and a fully non-symmetric functional-constraint model, whereas cyclopentadiene, nitrobenzene, azulene, and norcamphor probe cyclic, planar aromatic, polycyclic, and bridged cases. Across these benchmarks the generalized-internal-coordinate and symmetry-Cartesian refinements give the same numerical ranks, effective dimensionalities, and rotational residuals, while final structural parameters and uncertainties remain chemically interpretable.

# 1 Introduction

Accurate equilibrium molecular structures are central to rotational spectroscopy, thermochemistry, force-field development, and the validation of electronic-structure methods.[1,2] More generally, structure determination from experimental observables is a coordinate-dependent inverse problem. The data constrain a Cartesian geometry, but the fitted variables determine conditioning, constraints, correlations, and propagated uncertainties. Rotational spectroscopy is a stringent example because rotational constants probe the molecular mass distribution with high precision, whereas semiexperimental ($r_e^{\mathrm{SE}}$) structures require vibrational and sometimes electronic corrections to convert ground-state constants into equilibrium information.[3–6]

Modern semiexperimental and rotational-structure fitting programs and workflows, including the `MSR` molecular-structure-refinement program and the `STRFIT` structure-fitting program distributed with `PROSPE`,[7–9] have established robust statistical principles: fit moments of inertia or otherwise well-conditioned observables, include theoretical predicates when experimental information is incomplete, and propagate the covariance matrix to derived structural parameters.[10] The remaining bottleneck is the coordinate model rather than the least-squares formalism. This issue is easily hidden in small acyclic molecules, where a single Z-matrix reference tree can represent the important distances and angles without serious ambiguity.

Recent automated workflows have substantially extended the practical reach of semiexperimental refinement by coupling computed starting geometries, efficient second-order vibrational perturbation theory (VPT2)-based vibrational corrections, and reduced-dimensionality refinements in a largely black-box pipeline.[11] `SEfit` uses such computed geometries and correction data as upstream inputs and replaces the remaining user-defined Z-matrix layer with a topology-driven coordinate engine.

The same construction is not unique for rings, fused systems, and bridged frameworks. A Z-matrix is a tree representation of a molecular graph, whereas many molecular skeletons are not trees. Closing a ring therefore requires that at least one topological distance, angle, or dihedral be derived from the remaining variables rather than treated on the same footing. Different atom orders or reference paths then define different statistical models: one bond may be an adjustable parameter in one Z-matrix and a derived quantity in another, even though both describe the same Cartesian structure. Symmetry relationships and hydrogen constraints can compensate for this non-uniqueness, but only if they are introduced manually and consistently.

This work replaces the user-defined coordinate model by an automatic topology-driven layer. Starting from Cartesian coordinates, the coordinate engine builds primitive internal coordinates, removes redundancies, assigns symmetry, and supplies `SEfit` with the totally symmetric active space. Two coordinate choices are available. The default is a symmetry-adapted GIC basis constructed within homogeneous coordinate families, so stretches, bends, torsions, out-of-plane motions, and ring coordinates are not mixed indiscriminately. The alternative is a symmetry-adapted Cartesian basis obtained by projecting out translations and rotations and then applying the same point-group selection. Both are black-box topology-derived non-redundant coordinate models: every atom enters through the same Cartesian topology, no traversal path is selected, and no local structural parameter is privileged by construction.

The chemically meaningful interface remains internal-coordinate based. Constraints, predicates, and parameter classes are defined as primitive internal-coordinate relations and then projected onto whichever active basis is used. This is essential for the Cartesian option: a fixed C–H bond, an X–Y–H angle, or a local hydrogen frame is simple in primitive internal coordinates, whereas an equivalent Cartesian constraint is origin- and orientation-dependent, nonlocal, and generally not chemically transparent. The projection therefore uses the Wilson matrix or the corresponding primitive derivative operator[12] rather than asking the user to define Cartesian component constraints.

The rotational semiexperimental problem is used here as a demanding test of a broader question: which coordinates provide the most robust route from experimental observables to molecular structures? Glycolaldehyde represents a standard case where conventional Z-matrix methods are already adequate. Glycine I and II, cyclopentadiene, nitrobenzene, azulene, and norcamphor are deliberately less forgiving benchmarks because they require incomplete isotopic information, simple and coupled functional structural constraints, ring closures, planar-component selection, fused or bridged topology, and nontrivial constraints. The results show that automatic GICs and symmetry-adapted Cartesians lead to the same effective spaces and residuals, while final bond lengths, angles, dihedrals, and their uncertainties are reported uniformly from the optimized Cartesian geometry.

# 2 Theory and Method

## 2.1 Overview of the SEfit Workflow

The present framework has three layers. The first is the chemical interface: Cartesian geometry, topology, primitive internal coordinates, and any constraints, predicates, or parameter classes. The second is the active coordinate basis, chosen either as symmetry-adapted GICs or as symmetry-adapted Cartesian displacements. The third is the `SEfit` least-squares backend, which is independent of how the active basis was constructed.

This separation is important. The user specifies chemically meaningful relations in primitive internal coordinates, while the numerical solver sees a non-redundant totally symmetric active space. The final output is again chemical: optimized Cartesian coordinates, primitive bond lengths, angles, dihedrals, and propagated uncertainties.

The starting Cartesian geometry can be supplied by experiment, by any quantum chemical model, or by an automated composite-scheme generator. These starting geometries can also provide reference values for hard constraints, predicates, and local hydrogen frames when the isotopic data are insufficient to determine all hydrogen coordinates independently. The coordinate engine keeps the covalent topology used for primitive coordinates, ring detection, redundancy removal, and symmetry adaptation separate from any auxiliary reference information used to define constraints or predicates.

The overall architecture is summarized in Figure 1.

This separation between chemical specification, active coordinate basis, and least-squares refinement allows the same infrastructure to support both semiexperimental structure determination and other coordinate-dependent modules.

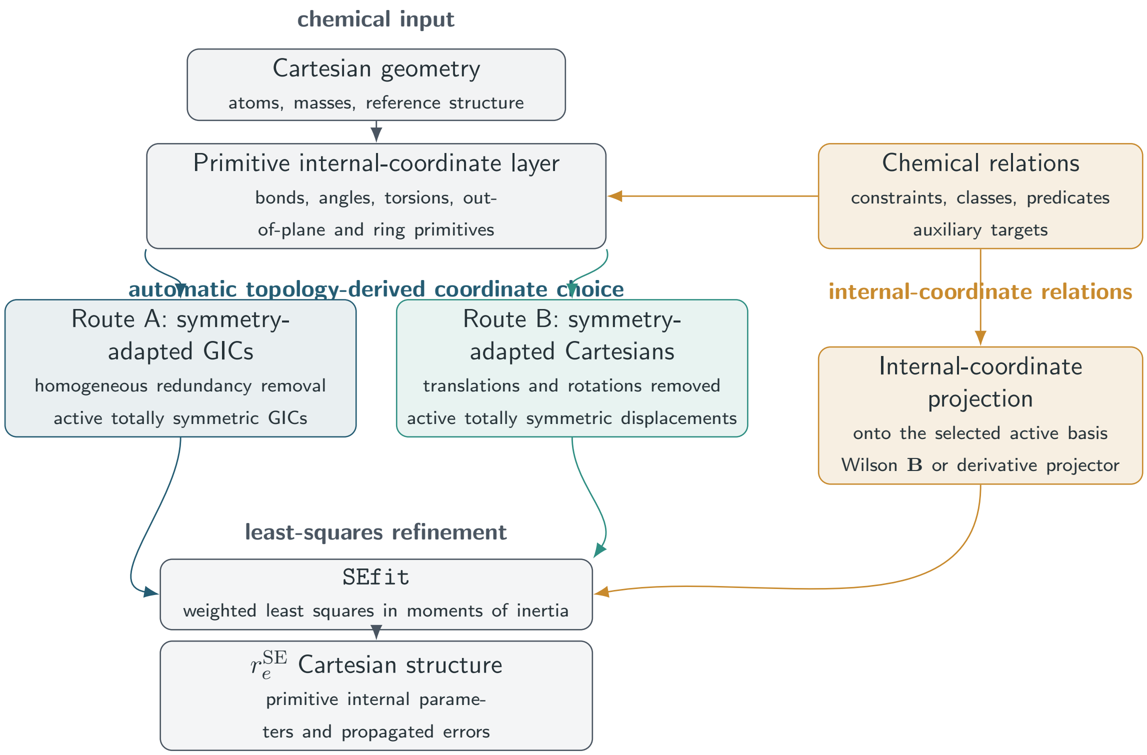


Figure 1: Architecture of the automatic `SEfit` workflow. Cartesian geometry defines topology and primitive internal coordinates once. The active least-squares space is then chosen as either Route A, a non-redundant symmetry-adapted GIC basis, or Route B, a symmetry-adapted Cartesian basis with translations and rotations removed. Constraints, parameter classes, and predicates remain internal-coordinate objects and are projected onto either active basis before refinement. The final Cartesian structure and the propagated errors of primitive internal parameters are generated by the same reporting layer. Auxiliary model information is handled in the chemical-relation layer and is not inserted into the covalent graph used to generate GICs.

## 2.2 Primitive Coordinate Generation

The coordinate construction starts from a Cartesian molecular geometry. The coordinate engine first perceives the covalent graph, including bonded atom pairs and ring connectivity. This graph is the only topology used for primitive-coordinate generation, ring detection, redundancy removal, and symmetry adaptation.

The graph is then expanded into a deliberately redundant primitive set: stretches, valence angles, torsions, out-of-plane deformations, linear bends, and ring-specific primitives. Redundancy at this stage is useful because it preserves locality and avoids forcing a premature choice of which primitive coordinates should become fitted variables.

Auxiliary reference information is kept outside the covalent graph. It may define constraints, predicates, or starting-geometry corrections, but it is not inserted into primitive GIC generation or ring detection.

For cyclic systems, ring coordinates are generated from endocyclic torsions and from analogous combinations of endocyclic valence angles. Unlike Cremer–Pople variables[13] or manually defined ring coordinates, these primitives remain within the generalized-internal-coordinate formalism and are available for monocyclic, fused-ring, bridged, and polycyclic systems.

This primitive layer is also the user-facing chemical interface. Constraints, predicates, and parameter classes are expressed here and remain independent of the non-redundant active representation generated later.

## 2.3 Coordinates for Parameter Refinement

Refinement coordinates must define statistical parameters, not merely a convenient optimization path. The normal matrix must have a controlled rank, the covariance matrix must be interpretable, and the resulting uncertainty must be propagated to reported bond lengths, angles, dihedrals, and other derived quantities. The active space is therefore required to be non-redundant, symmetry consistent, and compatible with chemically meaningful constraints.

This requirement exposes the non-uniqueness of Z-matrices. A Z-matrix selects a rooted traversal of the molecular graph. For rings, fused systems, and bridged skeletons, no traversal can make all chemically equivalent edges primary at the same time; at least one ring-closing distance or related quantity is derived. The fitted covariance is then attached to the chosen coordinate model as well as to the molecule.

Two active spaces are used here. In the GIC route, all topologically implied primitive internals are generated first and then reduced and symmetry adapted within homogeneous families, so stretches, bends, torsions, out-of-plane motions, and ring coordinates do not replace one another during rank reduction.[14–17] In the Cartesian route, translations and rotations are removed directly from the Cartesian displacement space and the totally symmetric block is retained. Both routes start from the same Cartesian geometry, use the same primitive-coordinate constraint layer, and report the same final structural observables with propagated uncertainties.

Symmetry both restricts the active variables to the totally symmetric representation and completes primitive constraints over equivalent orbits. Non-totally-symmetric coordinates remain available for diagnostics but are inactive in an equilibrium refinement.

## 2.4 Non-Redundant Symmetry-Adapted GICs

The primitive-coordinate set generated from topology is chemically intuitive but redundant. Linear dependencies from graph connectivity, ring closure, and symmetry are removed before refinement so that the normal equations and covariance matrix are defined in an independent coordinate space.

Let $\mathbf{x} \in \mathbb{R}^{3N}$ be the Cartesian coordinate vector and let

$$\mathbf{s}(\mathbf{x}) = (s_1(\mathbf{x}), s_2(\mathbf{x}), \ldots, s_m(\mathbf{x}))^T \tag{1}$$

collect the primitive internal coordinates. At a reference geometry $\mathbf{x}_0$, primitive displacements and Cartesian displacements are related to first order by the Wilson matrix[12]

$$\Delta\mathbf{s} = \mathbf{B}_s\Delta\mathbf{x}, \qquad \mathbf{B}_s = \left.\frac{\partial\mathbf{s}}{\partial\mathbf{x}}\right|_{\mathbf{x}_0}. \tag{2}$$

The primitive vector is partitioned into homogeneous families,

$$\mathbf{s} = \left(\mathbf{s}_{\mathrm{str}}, \mathbf{s}_{\mathrm{bend}}, \mathbf{s}_{\mathrm{tor}}, \mathbf{s}_{\mathrm{oop}}, \mathbf{s}_{\mathrm{ring}}, \ldots\right), \tag{3}$$

Reduction to non-redundant GICs precedes symmetry adaptation and is performed with a rank-revealing transformation of the full primitive set. Blocks are processed in fixed topological order with common singular-value thresholds. For each coordinate family $\alpha$, the Wilson block is

$$\mathbf{B}_{s_\alpha} = \frac{\partial\mathbf{s}_\alpha}{\partial\mathbf{x}}. \tag{4}$$

A singular-value decomposition,[18]

$$\mathbf{B}_{s_\alpha} = \mathbf{U}_\alpha\mathbf{\Sigma}_\alpha\mathbf{V}_\alpha^T, \tag{5}$$

identifies the independent primitive-coordinate differential space. Retained left-singular vectors define the row-orthonormal transformation

$$\mathbf{R}_\alpha = \mathbf{U}_{\alpha,r}^T, \tag{6}$$

where the subscript $r$ denotes the retained singular subspace. The non-redundant coordinates for family $\alpha$ are then

$$\Delta\mathbf{q}_\alpha = \mathbf{R}_\alpha\Delta\mathbf{s}_\alpha, \qquad \mathbf{B}_{q_\alpha} = \mathbf{R}_\alpha\mathbf{B}_{s_\alpha}. \tag{7}$$

The number of retained coordinates is the rank of $\mathbf{B}_{s_\alpha}$. If the stretching block is already full rank, primitive stretches are retained unchanged; other blocks are reduced only within the same physical family.

Stacking the blocks gives

$$\Delta\mathbf{q} = \mathbf{R}\Delta\mathbf{s}, \qquad \mathbf{B}_q = \mathbf{R}\mathbf{B}_s. \tag{8}$$

The block-diagonal structure of $\mathbf{R}$ preserves chemical homogeneity: rank reduction cannot use one coordinate family to replace another. After the family-wise reduction, $\mathbf{B}_q$ is checked against the expected vibrational rank. Any residual dependencies are pruned with the family blocks kept explicit and deterministic topological tie breaking. Molecular symmetry is applied only after this unsymmetrized non-redundant basis has been fixed.

Symmetry is then applied to the non-redundant GIC space. For each operation $g$ of the detected point group, the coordinate engine constructs the induced operation $\mathbf{D}_q(g)$ on $\Delta\mathbf{q}$ from the atom permutation, Cartesian rotation, and primitive-coordinate transformation. The projector onto irreducible representation $\Gamma$ is

$$\mathbf{P}_\Gamma^q = \frac{d_\Gamma}{|G|}\sum_{g\in G}\chi_\Gamma(g)\mathbf{D}_q(g), \tag{9}$$

where $d_\Gamma$ and $\chi_\Gamma$ are the dimension and character of $\Gamma$. Orthonormal columns of $\mathbf{S}_\Gamma$ span the range of this projector, and the symmetry-adapted GICs are

$$\Delta\mathbf{Q}_\Gamma = \mathbf{S}_\Gamma^T\Delta\mathbf{q}. \tag{10}$$

Only the totally symmetric block is active in an equilibrium refinement,

$$\Delta\mathbf{Q} \equiv \Delta\mathbf{Q}_{A_1} = \mathbf{S}_{A_1}^T\mathbf{R}\Delta\mathbf{s} = \mathbf{B}_Q\Delta\mathbf{x}, \tag{11}$$

with

$$\mathbf{B}_Q = \mathbf{S}_{A_1}^T \mathbf{R} \mathbf{B}_s. \tag{12}$$

Cartesian displacements associated with GIC corrections use the rank-controlled minimum-norm solution

$$\Delta \mathbf{x} = \mathbf{B}_Q^+ \Delta \mathbf{Q}, \tag{13}$$

where the Moore–Penrose pseudoinverse uses explicit singular-value thresholds and topology validation after accepted steps. For geometry corrections from target internal coordinates, the same derivative machinery can be used in weighted form by minimizing

$$\left\| \mathbf{W}_s^{1/2} \left( \mathbf{B}_t \Delta \mathbf{x} - \Delta \mathbf{t} \right) \right\|^2, \tag{14}$$

where $\mathbf{t}$ contains requested internal-coordinate changes or auxiliary reference targets. The diagonal metric $\mathbf{W}_s$ assigns different stiffness to coordinate families; auxiliary targets are not part of the covalent GIC active space unless a dedicated block is defined.

## 2.5 Symmetry-Adapted Cartesian Coordinates

The same least-squares machinery can also operate in a symmetry-adapted Cartesian displacement basis. This gives a direct Cartesian update route while retaining the same symmetry selection, constraint projection, and covariance propagation used for GICs.

Starting from the reference geometry, external translations and rotations are removed. Let $\mathbf{E} \in \mathbb{R}^{3N \times n_{\text{ext}}}$ collect these vectors, with $n_{\text{ext}} = 6$ for a nonlinear molecule and $n_{\text{ext}} = 5$ for a linear one. Rotational columns are generated from infinitesimal rotations around the center of mass,

$$\Delta \mathbf{r}_i = \boldsymbol{\omega} \times \left( \mathbf{r}_i - \mathbf{r}_{\text{cm}} \right), \tag{15}$$

for each atom $i$. The columns of $\mathbf{E}$ are orthonormalized to give $\mathbf{Z}_{\text{ext}}$, and the internal Cartesian projector is

$$\mathbf{P}_{\text{vib}} = \mathbf{I}_{3N} - \mathbf{Z}_{\text{ext}}\mathbf{Z}_{\text{ext}}^T. \tag{16}$$

This projector defines the non-redundant Cartesian displacement space before symmetry classification.

For each irreducible representation $\Gamma$ of the automatically detected point group, a Cartesian symmetry projector is constructed directly in the $3N$-dimensional displacement space as

$$\mathbf{P}_{\Gamma}^{x} = \frac{d_{\Gamma}}{|G|}\sum_{g\in G}\chi_{\Gamma}(g)\mathbf{R}(g), \tag{17}$$

where $\mathbf{R}(g)$ is the Cartesian operation, including atom permutation, associated with the symmetry operation $g$. Orthonormal columns spanning $\mathbf{P}_{\text{vib}}\mathbf{P}_{\Gamma}^{x}\mathbf{P}_{\text{vib}}$ define the symmetry-adapted Cartesian basis,

$$\text{range}\left(\mathbf{P}_{\text{vib}}\mathbf{P}_{\Gamma}^{x}\mathbf{P}_{\text{vib}}\right) = \text{span}\left(\mathbf{C}_{\Gamma}\right). \tag{18}$$

The columns of $\mathbf{C}_{\Gamma}$ are orthonormal displacement directions, not force-field normal modes. Only the totally symmetric block is used,

$$\Delta\mathbf{x} = \mathbf{C}_{A_1}\Delta\mathbf{q}_{\text{SC}}, \tag{19}$$

where $\Delta\mathbf{q}_{\text{SC}}$ are amplitudes along the totally symmetric Cartesian directions. Primitive variations induced by these amplitudes are

$$\Delta\mathbf{s} = \mathbf{B}_s\mathbf{C}_{A_1}\Delta\mathbf{q}_{\text{SC}}. \tag{20}$$

This links the Cartesian active space to the same primitive-coordinate interface used by

GICs. Bonds, angles, local hydrogen constraints, and Gaussian-style `ModRedundant` freezes are still specified as internal relations and projected by $\mathbf{B}_s\mathbf{C}_{A_1}$. Direct Cartesian constraints are avoided because a fixed bond or angle is a nonlinear, orientation-free relation distributed over several Cartesian components. Final bond lengths, valence angles, dihedrals, and uncertainties are evaluated from the optimized Cartesian geometry, as in the GIC route.

## 2.6 Constraints, Predicates, and Parameter Classes

User-defined structural relations are specified in primitive internal coordinates,

$$\mathbf{s} = (s_1, s_2, \ldots, s_m)^T, \tag{21}$$

not in GIC amplitudes or Cartesian components. The same primitive relation is projected onto either active space. Let $\Delta\mathbf{a}$ be the active correction vector. For GICs, $\Delta\mathbf{a} = \Delta\mathbf{Q}$ and

$$\Delta\mathbf{x} = \mathbf{T}_A\Delta\mathbf{a}, \qquad \mathbf{T}_A = \mathbf{B}_Q^+. \tag{22}$$

For symmetry Cartesians, $\Delta\mathbf{a} = \Delta\mathbf{q}_{\mathrm{SC}}$ and

$$\Delta\mathbf{x} = \mathbf{T}_A\Delta\mathbf{a}, \qquad \mathbf{T}_A = \mathbf{C}_{A_1}. \tag{23}$$

Primitive corrections induced by any active coordinate model are therefore

$$\Delta\mathbf{s} = \mathbf{B}_s\mathbf{T}_A\Delta\mathbf{a}. \tag{24}$$

This projector is used for constraints, predicates, classes, and error propagation.

**Hard constraints.** Hard constraints are exact equations imposed throughout the refinement:

$$s_i = s_i^0, \tag{25}$$

equalities,

$$s_i - s_j = 0, \tag{26}$$

or general linearized functions,

$$\mathbf{c}(\mathbf{s}) = \mathbf{d}. \tag{27}$$

At a trial geometry the corresponding first-order condition is

$$\mathbf{C}_s \Delta\mathbf{s} = \mathbf{b}_c, \qquad \mathbf{C}_s = \left.\frac{\partial \mathbf{c}}{\partial \mathbf{s}}\right|_{\mathbf{s}_0}, \qquad \mathbf{b}_c = \mathbf{d} - \mathbf{c}(\mathbf{s}_0). \tag{28}$$

Projection to the active space gives

$$\mathbf{C}_A \Delta\mathbf{a} = \mathbf{b}_c, \qquad \mathbf{C}_A = \mathbf{C}_s \mathbf{B}_s \mathbf{T}_A. \tag{29}$$

The solver keeps all compatible motions by writing the allowed correction as

$$\Delta\mathbf{a} = \Delta\mathbf{a}_0 + \mathbf{N}\Delta\mathbf{p}, \qquad \mathbf{C}_A \mathbf{N} = \mathbf{0}, \tag{30}$$

where $\Delta\mathbf{a}_0$ is a minimum-norm particular solution and $\mathbf{N}$ spans the null space. The least-squares problem is solved only for $\Delta\mathbf{p}$.

Symmetry-equivalent primitives are constrained as complete orbits before the null-space projector is formed. Gaussian-style `ModRedundant` freezes and generalized-internal-coordinate expressions are interpreted in this primitive layer. A line `Name=Expression` defines a reusable coordinate; `Frozen,Value=` turns it into a hard equation. Names can combine stretches, angles, dihedrals, out-of-plane coordinates, Cartesian components, or final `GIC`$n$ labels. If `Value=` is omitted, the starting-geometry value is frozen; tabulated structural information therefore uses explicit `Value=` targets.

Reduced-dimensionality refinements use the same projection. Hydrogen coordinates supported by isotopic data remain refinable, whereas unsupported local hydrogen frames can be frozen or tied without over-constraining the heavy-atom skeleton. Additional model-dependent geometry information is kept outside the covalent topology used for GIC generation.

**Predicate observations.** Predicates are soft structural information. They add weighted residuals rather than removing degrees of freedom.[10] A primitive predicate has the form

$$f(\mathbf{s}) = f^{\text{ref}} \pm \sigma_f, \tag{31}$$

and contributes

$$r_f = \frac{f(\mathbf{s}) - f^{\text{ref}}}{\sigma_f} \tag{32}$$

to the residual vector. Its Jacobian with respect to the independent least-squares variables is

$$\mathbf{J}_f = \frac{1}{\sigma_f} \frac{\partial f}{\partial \mathbf{s}} \mathbf{B}_s \mathbf{T}_A \mathbf{N}. \tag{33}$$

Their uncertainties control their statistical weight.

**Parameter classes.** Parameter classes make several primitive quantities share one correction. If class $c$ contains primitives $i \in c$, the allowed correction is

$$\Delta s_i = \Delta p_c, \qquad i \in c. \tag{34}$$

Class definitions are converted into linear relations or into a Jacobian compression before the normal equations are solved. The operation applies to corrections rather than final values, so chemically related quantities may retain different reference values while sharing a fitted offset. After convergence, the covariance is expanded back to the reported primitive and topological parameters.

## 2.7 Semiexperimental Observables

For each isotopologue $s$, the measured ground-state rotational constants are converted into equilibrium rotational constants by removing vibrational and, when necessary, electronic contributions,[3–6]

$$B_{\alpha,e}^{(s)} = B_{\alpha,0}^{(s)} - \Delta B_{\alpha,\mathrm{vib}}^{(s)} - \Delta B_{\alpha,\mathrm{elec}}^{(s)}, \qquad \alpha \in \{a, b, c\}. \tag{35}$$

The correction terms may come from any electronic-structure or composite protocol and are supplied through the isotopologue data file together with uncertainties, masses, conventions, and provenance.[11]

Although rotational constants may be fitted directly, `SEfit` uses principal moments of inertia as the default observables,

$$I_\alpha^{(s)} = \frac{K}{B_{\alpha,e}^{(s)}}, \tag{36}$$

where $K$ is the standard rotational conversion constant.

Moments of inertia give a smoother geometry dependence and avoid reciprocal amplification of small constant variations.

Rotational constants can still be fitted for diagnostics. For planar molecules, however, only two principal components are independent. If the molecular plane is the $ab$ plane,

$$I_c = I_a + I_b, \tag{37}$$

Fitting all three components would duplicate one constraint. `SEfit` therefore uses one pair, $AB$, $AC$, or $BC$, and reports the omitted component as a diagnostic. Planarity is recognized from the singular values of the centered Cartesian geometry. If the user does not choose a pair, the code tests all admissible pairs, keeps the largest numerical rank, and among rank-equivalent choices minimizes the propagated instability of the omitted moment from the planar identity. The Jacobian condition number is only the final tie-breaker; forced $ABC$ planar fits are rejected as redundant.

The complete experimental data set consists of the selected observables for all isotopologues and their uncertainties. These values form the experimental part of the least-squares objective described below.

## 2.8 Least-Squares Refinement

The semiexperimental refinement is formulated as a weighted nonlinear least-squares problem. The active coordinate vector $\mathbf{a}$ is either the totally symmetric GIC vector or the totally symmetric Cartesian displacement amplitude vector. After hard constraints and parameter classes have been projected, the actual independent least-squares variables are collected in $\mathbf{p}$.

The residual vector is

$$\mathbf{r} = \mathbf{y}^{\text{obs}} - \mathbf{y}^{\text{calc}}(\mathbf{p}), \tag{38}$$

where $\mathbf{y}$ collects all experimental observables and predicate observations included in the refinement.

The objective function is

$$\chi^2 = \mathbf{r}^T \mathbf{W} \mathbf{r}, \tag{39}$$

where $\mathbf{W}$ is the diagonal weight matrix constructed from the inverse variances of the experimental and predicate observations. When robust fitting is requested, the experimental part of $\mathbf{W}$ is modified by an iteratively reweighted loss applied to complete isotopologue blocks rather than to individual rotational constants. Thus all selected components of a suspect isotopologue are attenuated together, whereas quantum-chemical predicates retain their assigned statistical weights.

The optimization is performed using a weighted Levenberg–Marquardt procedure,

$$\left(\mathbf{J}^T\mathbf{W}\mathbf{J} + \lambda\mathbf{I}\right)\Delta\mathbf{p} = \mathbf{J}^T\mathbf{W}\mathbf{r}, \tag{40}$$

where $\mathbf{J}$ is the Jacobian matrix of the observables with respect to the independent variables $\mathbf{p}$ and $\lambda$ is the damping parameter. Equation (40) is not solved in the implementation by forming the normal equations. The actual step is obtained from the column-scaled augmented system

$$\begin{bmatrix} \mathbf{W}^{1/2}\mathbf{J}\mathbf{S} \\ \lambda^{1/2}\mathbf{I} \end{bmatrix} \Delta\mathbf{u} = \begin{bmatrix} \mathbf{W}^{1/2}\mathbf{r} \\ \mathbf{0} \end{bmatrix}, \qquad \Delta\mathbf{p} = \mathbf{S}\Delta\mathbf{u},$$

using rank-revealing singular-value-decomposition (SVD) and QR-factorization linear algebra.[18] The diagonal scaling matrix $\mathbf{S}$ is updated from the current weighted Jacobian column norms, on top of the chemical coordinate-family scaling. This prevents weakly observed variables or mixed units from controlling the trust-region step and avoids the condition-number squaring associated with explicit normal equations.

The Jacobian is evaluated analytically whenever possible. Independent coordinate corrections are transformed into active-coordinate corrections, then into Cartesian displacements through the projector $\mathbf{T}_A\mathbf{N}$ described above. This gives analytic derivatives of moments of inertia and rotational constants with respect to the variables that are actually fitted.

The solver uses a trust-region Levenberg–Marquardt strategy based on the predicted and actual reduction of the weighted objective function.[19] Trial line-search points are evaluated with a frozen coordinate model. Every accepted step is then validated by the coordinate-definition service against the original topology, point group, irreducible-representation counts, and coordinate-family counts. Steps that would change this GIC signature are rejected and the trust radius is reduced. The Cartesian–GIC projector is refreshed analytically when needed and otherwise updated by a secant correction, avoiding unnecessary coordinate rebuilds without accepting topology-changing steps. The same numerical kernel is mirrored in the Fortran77 backend: dynamic column scaling, rank-revealing QR solution of the augmented Levenberg–Marquardt system, and grouped robust weights are available without importing legacy Gaussian code.

Convergence requires both the parameter corrections and the objective-function reduction to fall below predefined thresholds. Trial steps that increase the objective are rejected automatically, and the damping parameter is adjusted adaptively. For validation, `SEfit` can perform exact leave-one-isotopologue-out refits. Each isotopologue is removed in turn, the geometry is refitted with the same coordinate and constraint model, and the omitted rotational constants are predicted from the resulting structure. These diagnostics are not part of the objective function; they identify outlier isotopologues and quantify how strongly individual substitutions control the fitted geometry. The same philosophy is used for numerical warnings: reduced rank, small singular values, strongly downweighted isotopologues, ill-conditioned planar component pairs, and unusually large propagated coordinate uncertainties are reported explicitly, but they do not redefine the coordinate model or the least-squares objective.

Following convergence, the covariance matrix in the independent fitted variables is obtained from

$$\mathbf{C}_p = \sigma_r^2 \left(\mathbf{J}^T\mathbf{W}\mathbf{J}\right)^{-1}, \tag{41}$$

where $\sigma_r^2$ is the reduced residual variance.

## 2.9 Error Propagation

Uncertainty propagation follows the same chain of transformations used for the geometry update. If $\mathbf{N}$ is the null-space basis of the hard constraints and class reductions, the covariance in the active coordinate space is

$$\mathbf{C}_a = \mathbf{N}\mathbf{C}_p\mathbf{N}^T. \tag{42}$$

The corresponding Cartesian covariance is

$$\mathbf{C}_x = \mathbf{T}_A\mathbf{C}_a\mathbf{T}_A^T. \tag{43}$$

For a GIC refinement, $\mathbf{T}_A = \mathbf{B}_Q^+$; for a Cartesian refinement, $\mathbf{T}_A = \mathbf{C}_{A_1}$. Therefore the same covariance propagation is used for both coordinate choices.

Any reported structural quantity is treated as a function of the optimized Cartesian geometry. For a vector of reported quantities $\mathbf{g}(\mathbf{x})$, which may contain bond lengths, valence angles, dihedral angles, out-of-plane coordinates, ring descriptors, or user-defined primitive functions, the covariance is

$$\mathbf{C}_g = \mathbf{G}_x\mathbf{C}_x\mathbf{G}_x^T, \qquad \mathbf{G}_x = \left.\frac{\partial \mathbf{g}}{\partial \mathbf{x}}\right|_{\mathbf{x}_{\mathrm{opt}}}. \tag{44}$$

The reported standard errors are the square roots of the diagonal elements of $\mathbf{C}_g$, while the off-diagonal elements provide correlations among derived structural parameters. Angular derivatives are evaluated in radians and converted to degrees only in the final report. Symmetry-equivalent reported quantities can either be listed separately with their covariance or condensed by the same linear propagation formula.

Hard constraints have no variance along the constrained directions because those directions are removed before the normal equations are solved. Predicate observations are different: their stated uncertainties enter the weight matrix and therefore contribute statistically to $\mathbf{C}_p$. This distinction is important when hydrogen parameters are held fixed because deuterated isotopologues are unavailable. A frozen local hydrogen frame is a constraint; a quantum-chemical C–H distance with an assigned uncertainty is a predicate. Only the latter contributes uncertainty through its assigned standard deviation.

The output therefore reports both standard errors and correlation diagnostics. Large correlations are not treated as failures by themselves, because they are expected whenever isotope substitutions do not isolate all local structural directions. They are used to identify underdetermined parameter combinations, to decide whether a primitive should be fixed or converted into a predicate, and to verify that reported derived quantities are propagated from the full covariance matrix rather than from independent one-parameter estimates. The same information is complemented by an SVD diagnostic of the final weighted Jacobian. Small singular values are reported together with the dominant coordinate combinations in the corresponding right singular vectors, so a rank deficiency can be traced to specific GICs, parameter classes, or symmetry-Cartesian directions instead of being summarized only by a condition number.

All constraints are also audited after projection. The output records the input primitive patterns, the symmetry-equivalent primitive constraints generated from them, the parameter classes, and the active working-coordinate labels actually matched by each entry. This is important for black-box refinements: the user does not construct the active coordinate basis manually, but can verify exactly which chemically stated restrictions were imposed on that basis.

The final structural tables are thus independent of the internal coordinate basis used during optimization. The fitted variables may be GIC amplitudes or Cartesian amplitudes, but bond lengths, angles, dihedrals, and ring parameters are always evaluated from the same optimized Cartesian geometry and receive errors from the same propagated covariance matrix.

After convergence, `SEfit` also evaluates the numerical Hessian of the weighted least-squares objective with respect to the independent variables that remain after symmetry selection, parameter-class reduction, and constraint projection. Diagonalization of this optimized-space Hessian provides the final stationary-point diagnostic: a positive spectrum identifies a local minimum in the fitted coordinate space, negative eigenvalues flag an unstable direction, and near-zero eigenvalues are reported as rank-deficient or statistically flat directions rather than hidden by the refinement. The benchmark tables therefore report the stationary-point classification, and, where useful, the smallest Hessian eigenvalue.

### 2.10 Kraitchman Diagnostic

For singly substituted isotopologues, `SEfit` computes substitution coordinates from the Kraitchman equations[20] and compares their absolute values with the fitted coordinates in the parent principal-axis frame. This diagnostic is intentionally separate from the fit: Kraitchman coordinates are useful assignment checks, but they do not retain coordinate signs and do not exploit the full correlated least-squares model.

## 3 Implementation and Reproducibility

The implementation is organized around two reusable services. The coordinate-definition service defines the molecular coordinate model from Cartesian atoms and nuclear charges, assigning point-group symmetry and irreducible representations when requested. A separate B-matrix service evaluates the same frozen GIC definition for any compatible Cartesian geometry. `SEfit` calls the definition service only at a fresh start and reuses the stored coordinate model during the least-squares iterations, rebuilding or updating the B matrix only when the current step requires it. The command-line interface and the graphical interface call these same services and expose the choice between GIC and symmetry-adapted Cartesian active spaces.

The standard input model contains Cartesian parent geometry, optional Gaussian-style primitive freeze records, and a separate isotopologue file with substitutions, ground-state rotational constants, vibrational and electronic corrections, uncertainties, and correction conventions. Legacy MSR-style input is accepted for reproducibility, including frozen variables marked by `#`, but it is converted immediately into the same Cartesian, primitive-constraint, and isotopologue representation. Z-matrices are therefore compatibility inputs, not working coordinates.

Each run writes a text report, final Cartesian coordinates, primitive internal parameters with propagated uncertainties, corrected/calculated rotational observables, residuals, covariance/correlation and influence diagnostics, SVD, constraint, and warning diagnostics, Hessian eigenvalues for the minimum check, Kraitchman diagnostics when applicable, and a machine-readable manifest. A checkpoint file stores the Cartesian geometry, active labels, damping/trust region state, and robust weights; restart calculations rebuild the GIC or symmetry-Cartesian definitions deterministically from those Cartesian coordinates. The benchmark tables in this article are generated from the versioned run comma-separated-value (CSV) files by a single regression command, so the manuscript values, test suite, and archived examples use the same numerical source. A versioned distribution of the `SEfit` source code, examples, and regression tests will be made available to reviewers and, upon publication, under the distribution model selected by the authors.

# 4 Relationship to Existing Approaches

`SEfit` combines established semiexperimental least-squares machinery[7,10,11] with an automatically generated coordinate layer. The statistical model uses moments of inertia, vibrational and electronic corrections, predicates, parameter classes, leverage and correlation diagnostics, robust weighting, and final minimum checks. The new element is not the observable model but the definition of the refinement variables.

Z-matrices remain compact for tree-like molecules, as illustrated below by glycolaldehyde.

Their limitations become visible for rings, fused systems, and bridged frameworks: a molecular graph with cycles cannot be represented by a single reference tree without making some chemically equivalent quantities primary and others derived. In a least-squares refinement this affects not only notation but also the covariance attached to reported parameters.

Generalized internal coordinates are a flexible alternative and are widely used in optimization and vibrational analysis.[12,14–17,21] Gaussian, for example, provides a powerful GIC language for coordinate combinations, constraints, scans, and conditional definitions.[15,21] In those workflows the generalized coordinates are usually a user-defined layer. Here they are generated from topology-based primitive coordinates, reduced to a non-redundant space, assigned to irreducible representations, and accessed through primitive-coordinate constraints. Kisiel's `STRFIT`/PROSPE route[8] provides another important reference point for rotational-structure fitting, but the structural model is again supplied by the user and may leave derived ring quantities outside the primary fitted covariance matrix.[9]

The practical distinction is summarized in Table 1. `SEfit` retains the usual semiexperimental inputs and diagnostics while moving coordinate construction, symmetry completion, and covariance propagation to a common topology-derived layer.

# 5 Validation and Representative Applications

The validation set was chosen to separate standard and non-standard coordinate problems. Glycolaldehyde is a small acyclic molecule for which a traditional Z-matrix is straightforward; it verifies that the automatic procedure does not lose efficiency in a simple case. Glycine I and II test reduced-dimensionality refinements at two levels of complexity: Gly-I retains $C_s$ symmetry and uses four simple structural constraints, whereas Gly-II is treated without symmetry and uses coupled functional constraints. Cyclopentadiene, nitrobenzene, azulene, and norcamphor are more demanding because rings, planar rotational redundancy, fused topology, bridged connectivity, and missing deuterium substitutions force conventional workflows

Table 1: Operational comparison of representative structure-refinement routes. The table emphasizes coordinate construction and uncertainty reporting, not the statistical quality of individual refinements.

| Feature | Z-matrix semiexperimental workflows | User-defined GIC or `STRFIT`/PROSPE workflows | `SEfit` |
|---|---|---|---|
| Primary geometry model | User-selected internal coordinates | User-supplied structural or generalized-coordinate model | Cartesian input with automatic topology perception |
| Ring and fused-ring treatment | Manual reference-tree choices | Manual parameter or coordinate selection | Automatic ring primitives and non-redundant active spaces |
| Symmetry handling | Specified in the chosen coordinate model | Defined by the supplied structural model | Automatic point-group detection and symmetry completion |
| Active variables | User-selected structural parameters | User-selected fitted parameters or GICs | Totally symmetric GICs or symmetry Cartesians |
| Constraints and classes | Expressed in the selected model | Expressed in the supplied model | Expressed as primitive internals and projected automatically |
| Missing substitution data | Reduced-dimensionality choices in the model | Fixed or selected parameters | Constraints, predicates, classes, or symmetry-complete local frames |
| Derived structural uncertainties | Propagated according to the selected coordinate model | Propagated according to the supplied fitted-parameter model | Propagated from the optimized Cartesian structure |
| Diagnostics | Established least-squares diagnostics | Diagnostics provided by the selected program and workflow | Rank, conditioning, robust weights, correlations, and leave-one-out checks |

to choose derived coordinates, fixed hydrogen models, or manually coupled parameters.

No molecule-specific coordinate code was introduced. In every benchmark the input structure was Cartesian. The coordinate engine generated the coordinate model, `SEfit` selected the totally symmetric active space, and any constraints were supplied as primitive internal-coordinate relations. The same constraint definitions were used when the active basis was changed from GICs to symmetry-adapted Cartesians.

The regression suite additionally checks GIC generation for pyridine, uracil, guanine, water, cyclopentadiene, nitrobenzene, and azulene. These tests verify point-group assignment, non-redundant coordinate counts, and totally symmetric dimensions independently of the semiexperimental fit.

The atom numbering used in structural tables and diagnostic discussions is shown in Figure 2. The labels come directly from the Cartesian input order and the perceived topology, so they match the numbering used by the coordinate engine, `SEfit`, and the primitive-coordinate constraints.

Table 2: One-page summary of the representative `GIC_Forge/SEfit` semiexperimental refinements. "Model" is the working-coordinate model used for the main row. "Fit" gives the independent rotational components entering the normal equations; for moment-based fits, $AB$ denotes $(I_a, I_b)$. "GIC/A" gives the final non-redundant GIC count and the totally symmetric active count before constraint projection. "Eff./rank" gives the independent variables after primitive-constraint projection and the final numerical rank. Rotational residuals are RMS/maximum absolute values in MHz over all reported diagnostic constants.

| System | PG | Model | Fit | Iso. | GIC/A | Eff./rank | Constraints | Steps | RMS/max |
|---|---|---|---|---|---|---|---|---|---|
| Glycolaldehyde | $C_s$ | GIC | ABC | 10 | 18/12 | 12/12 | none | 3/0; min | 0.5640/1.7961 |
| Glycine I | $C_s$ | GIC | ABC | 5 | 24/15 | 11/11 | 4 function | 7/25; min | 0.0271/0.0565 |
| Glycine II | $C_1$ | GIC | ABC | 10 | 24/24 | 19/19 | 5 function | 16/30; min | 0.2491/0.6707 |
| Cyclopentadiene | $C_{2v}$ | Cart. | ABC | 14 | 27/10 | 10/10 | none | 10/0; min | 0.0635/0.2289 |
| Nitrobenzene | $C_{2v}$ | GIC | AB | 9 | 36/13 | 13/13 | 8 primitive | 4/0; pos. | 0.1480/0.2605 |
| Azulene | $C_{2v}$ | GIC | AB | 12 | 48/17 | 10/10 | 8 primitive | 8/28; min | 0.0731/0.2900 |
| Norcamphor | $C_1$ | GIC | ABC | 9 | 48/48 | 18/18 | 30 H primitive | 8/0; min | 0.0710/0.2091 |

Table 3: Planar-component diagnostics for the planar benchmarks. Only two principal moments are independent for a planar molecule; the selected pair marked by an asterisk is used in the normal equations, while all three rotational constants are recomputed after convergence. The residuals are RMS/maximum absolute MHz values over the complete diagnostic $A$, $B$, and $C$ constant set.

| System | Pair | Fitted moments | Rank | Cond. | Diagnostic RMS/max |
|---|---|---|---|---|---|
| Nitrobenzene | AB* | $(I_a, I_b)$ | 13 | 27464.4 | 0.1480/0.2605 |
| Nitrobenzene | AC | $(I_a, I_c)$ | 13 | 56007.7 | 0.2593/0.4597 |
| Nitrobenzene | BC | $(I_b, I_c)$ | 13 | 10623.3 | 2.4602/4.3693 |
| Azulene | AB* | $(I_a, I_b)$ | 10 | 956.7 | 0.0731/0.2900 |
| Azulene | AC | $(I_a, I_c)$ | 10 | 1780.0 | 0.1086/0.4222 |
| Azulene | BC | $(I_b, I_c)$ | 10 | 950.2 | 0.3232/0.7541 |

Table 3 makes the planar choice explicit. The selected $AB$ pair is not an arbitrary convention: it retains the two in-plane moments directly and leaves the perpendicular constant as a diagnostic. Forced $AC$ or $BC$ fits have the same rank but give larger complete rotational diagnostic residuals for both nitrobenzene and azulene. In Table 2, "pos." denotes a rank-deficient statistical Hessian with positive curvature in all retained fitted directions.

## 5.1 Coordinate Content of the Benchmark Refinements

The least-squares variables are not necessarily the topological quantities listed in the final structural tables. They are either symmetry-adapted GICs or symmetry-adapted Carte-

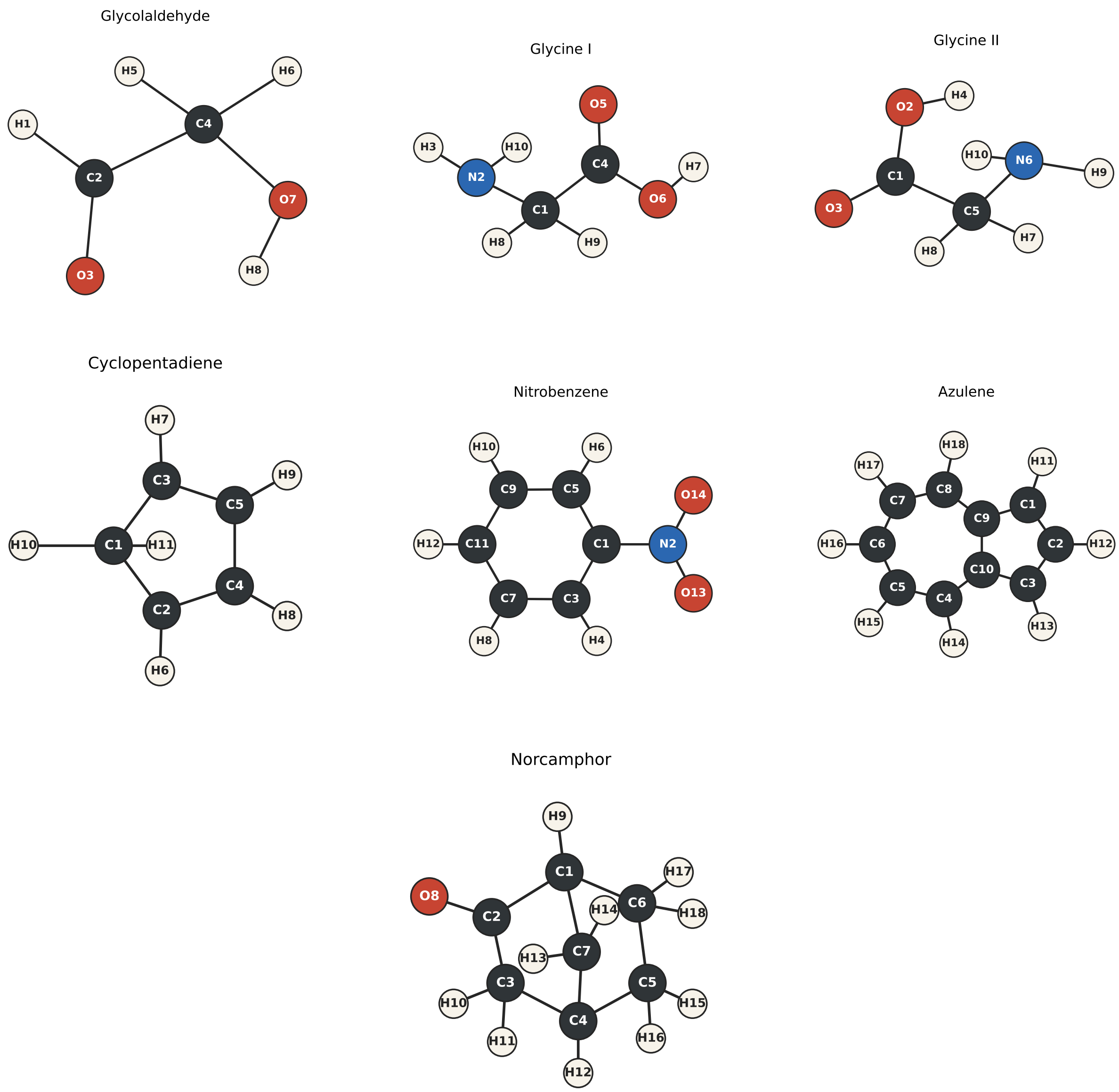


Figure 2: Atom numbering used for the glycolaldehyde, glycine I, glycine II, cyclopentadiene, nitrobenzene, azulene, and norcamphor semiexperimental refinements. The labels are derived from the Cartesian input order and are used consistently in the GIC definitions, primitive-coordinate constraints, and structural tables.

sian amplitudes. In both cases, reported bond lengths, angles, dihedrals, and out-of-plane quantities are evaluated after convergence from the optimized Cartesian geometry, with uncertainties propagated from the fitted covariance matrix.

The primitive layer contains the local coordinates implied by the perceived graph: stretches, valence angles, torsions, and ring deformation coordinates derived from endocyclic valence-angle and torsional primitives. This redundant set is reduced and symmetry-adapted in homogeneous blocks. Stretching GICs are therefore formed only from stretches, valence-angle GICs only from valence-angle primitives, and torsional or ring-puckering GICs only from torsional-type primitives. This prevents a bond-stretching correction from being replaced by a ring-puckering or torsional correction during rank reduction.

The coordinate counts in Table 4 refer to the covalent-topology GIC space. Auxiliary reference targets are deliberately not included in these counts, because they do not create additional ring or GIC primitives.

Table 4 summarizes the GIC coordinate content. For glycolaldehyde the coordinate problem is standard: the active $A'$ variables are stretches and bends. Glycine I is a $C_s$ constrained case, whereas glycine II and norcamphor are complementary $C_1$ cases: all GICs are formally totally symmetric, so dimensionality reduction comes from exact chemical constraints rather than symmetry. For nitrobenzene and azulene, planar aromatic symmetry leaves stretching and in-plane bending coordinates in the totally symmetric active space, whereas torsional and out-of-plane ring coordinates are retained in the complete coordinate report but excluded from the equilibrium refinement.

Primitive constraints reduce the active spaces only after the totally symmetric coordinates have been selected. In azulene, selected MSR-inspired C–H distances and valence angles reduce the 17 totally symmetric GICs to ten effective variables. In glycine I, four tabulated constraints freeze the O–H distance, C–O–H angle, H–N–H angle, and $NH_2$ wagging coordinate, reducing the 15-dimensional $A'$ space to 11 variables. In glycine II, five Gaussian-style functional constraints couple C–H and N–H differences, methylene rocking/twisting combi-

Table 4: Coordinate content of the benchmark refinements. Numbers in parentheses are the totally symmetric coordinates within each family. The active variables are the totally symmetric coordinates before any primitive-constraint projection; in $C_1$, all coordinates transform as $A$.

| System | Total | Stretching | Bending | Torsion/ring | Active totally symmetric character |
|---|---|---|---|---|---|
| Glycolaldehyde | 18 | 7 (6) | 8 (6) | 3 (0) | 6 stretches + 6 bends |
| Glycine I | 24 | 9 (7) | 10 (7) | 5 (1) | $C_s$ model; 4 functional constraints project 15 $A'$ variables to 11 |
| Glycine II | 24 | 9 (9) | 10 (10) | 5 (5) | all GICs in $C_1$; 5 functional constraints project to 19 variables |
| Cyclopentadiene | 27 | 11 (6) | 10 (4) | 6 (0) | 6 stretches + 4 bends |
| Nitrobenzene | 36 | 14 (8) | 11 (5) | 11 (0) | 8 stretches + 5 bends |
| Azulene | 48 | 19 (11) | 14 (6) | 15 (0) | 11 stretches + 6 bends |
| Norcamphor | 48 | 18 (18) | 25 (25) | 5 (5) | all GICs in $C_1$; 30 primitive H constraints project to 18 variables |

nations, and the average H–N–C–C torsion, reducing the 24-dimensional $C_1$ active space to 19 variables. In norcamphor, the absence of deuterium substitutions is handled by freezing one C–H stretch, one C–C–H angle, and one C–C–C–H torsion for each hydrogen, leaving an 18-dimensional heavy-skeleton refinement. The same chemical information is therefore used as in the published reduced-dimensionality models, but the least-squares variables are generated automatically rather than supplied as Z-matrix parameters.

## 5.2 Glycolaldehyde Small-Molecule Benchmark

Glycolaldehyde was included as a compact acyclic benchmark. It is not a case where Z-matrix methods are expected to fail; rather, it verifies that the automatic Cartesian/topology workflow reproduces a standard refinement without requiring any manual coordinate model. The input geometry was the published composite-scheme Cartesian structure reported in the Supporting Information of a recent rotational-spectroscopy benchmark, and the spectroscopic data consisted of the parent species, singly substituted $^{13}$C and $^{18}$O isotopologues, hydroxyl and carbon-bound deuterated species, and two doubly substituted deuterated species. The ground-state rotational constants were corrected with computed vibrational contributions using the additive convention $B_e = B_0 + \Delta B_{\mathrm{vib}}$.

The coordinate engine assigned $C_s$ symmetry and generated 18 final non-redundant GICs:

seven stretching, eight bending, and three torsional coordinates. Twelve transform as $A'$ and were active in the refinement; the $A''$ coordinates were retained in the complete report but excluded from the equilibrium fit. The refinement converged in three accepted steps without rejections, and the optimized-space Hessian was positive definite.

Table 5 gives the diagnostics. The rotational-constant RMS residual is larger than for the MSR-derived cyclopentadiene benchmark because the absolute residuals are dominated by the large $A$ constants, while the fitted moment-of-inertia residuals remain small. The largest rotational residual is 1.7961 MHz for the $A$ component of the $^{13}$C2 isotopologue. A hydrogen-local-frame frozen comparison also converges to a minimum but gives a larger RMS residual of 1.7893 MHz, showing that the deuterated data carry useful information on hydrogen positions. A sign check using subtractive vibrational corrections gives a much poorer RMS residual of 15.99 MHz, so the additive convention is retained.

Table 5: Glycolaldehyde small-molecule semiexperimental benchmark. Moments of inertia were used as fit observables; rotational residuals are reported in MHz.

| Quantity | Value |
|---|---:|
| Isotopologues | 10 |
| Rotational observables | 30 |
| Final non-redundant GICs | 18 |
| Active $A'$ GICs | 12 |
| Numerical rank | 12 |
| Accepted/rejected steps | 3/0 |
| Condition number | 347.82 |
| Weighted RMS in inertia space | $2.83 \times 10^{-3}$ |
| RMS rotational-constant residual | 0.5640 |
| Maximum rotational-constant residual | 1.7961 |
| Reduced $\chi^2$ | $1.34 \times 10^{-5}$ |
| Smallest Hessian eigenvalue | 4.3757 |
| Stationary point | minimum |

## 5.3 Glycine I and II Functional-Constraint Benchmarks

Glycine conformer I provides a compact constrained $C_s$ benchmark. The fit uses the parent, $^{13}$C(carboxyl), $^{13}$C(methylene), $CD_2$, and $^{15}$N isotopologues. The ground-state rotational constants and statistical weights are taken from Godfrey and Brown,[22] whereas the Gly-Ip zero-point corrections and the four structural constraints come from Kasalová et al.[23] The constraints are imposed directly in Gaussian-style syntax: the O–H distance, the C–O–H angle, the H–N–H scissor angle, and the $NH_2$ wagging out-of-plane coordinate. GICForge assigns $C_s$ symmetry, generates 24 final GICs, and selects 15 $A'$ coordinates. Projection of the four constraints leaves 11 independent fitted variables. The weighted fit reaches a minimum with seven accepted steps, condition number $1.02 \times 10^4$, complete rotational RMS residual 0.0271 MHz, maximum residual 0.0565 MHz, and a positive optimized-space Hessian. The complete rotational comparison and primitive structural parameters are reported in the Supporting Information.

Glycine conformer II is a more demanding reduced-dimensionality benchmark because the available isotopic information does not by itself determine a stable semiexperimental structure. Kasalová et al. showed that high-level all-electron cc-pVTZ CCSD(T) structural information and vibrational corrections are needed to obtain a meaningful Gly-IIn refinement from the microwave isotopologue set.[23] This makes Gly-II a useful test of whether model information can be supplied as chemically transparent constraints without reverting to a Z-matrix.

The present calculation starts from the non-planar DPCS3 Cartesian geometry and uses the 10-isotopologue Gly-IIn data set with additive zero-point vibrational corrections. No $C_s$ constraint is imposed: GICForge assigns the $C_1$ coordinate space, so all 24 final GICs are formally totally symmetric. The five reduced-dimensionality relations have the same functional form as the constraints used in the published refinement: a C–H difference, an N–H difference, two methylene angular combinations, and an average H–N–C–C torsion. They are supplied in Gaussian-style syntax as named coordinate definitions followed by `Frozen,Value=` equations, for example `RCH7=R(5,7)` and `DRCH(Frozen,Value=0.000091749296)=RCH8-RCH7`. Thus the constraints are functions of primitive coordinates, not fitted Z-matrix variables. Their target values were evaluated from the DPCS3 starting geometry.

Table 6 compares the DPCS3-derived targets with a run using the published Kasalová-type target values. The two fits have the same rank and essentially the same residual scale; the DPCS3 targets therefore reproduce the coupled-cluster constraint information closely enough for this benchmark while remaining generated by the same upstream Cartesian-geometry machinery used elsewhere in the workflow.

Table 6: Glycine II constraint-source comparison. Both calculations used GIC coordinates, moments of inertia as observables, and the same 10 corrected Gly-IIn rotational data sets. Rotational residuals are reported over the diagnostic $A$, $B$, and $C$ constants in MHz. The last column is the mean-square convention used for comparison with the published glycine refinement, not a second RMS value.

| Constraint target source | Rank | RMS | Max | $10^3\langle\Delta B^2\rangle$ |
|---|---|---|---|---|
| Kasalová-type values | 19 | 0.2344 | 0.8100 | 54.94 |
| DPCS3-derived values | 19 | 0.2491 | 0.6707 | 62.06 |

The final DPCS3-constrained refinement converges to a minimum with 19 effective variables, a condition number of $1.05 \times 10^4$, and a complete rotational RMS residual of 0.2491 MHz. The larger uncertainties are associated with the amino-hydrogen frame, as expected from the incomplete substitution pattern. Table 7 reports the final topological parameters and propagated standard errors. These quantities are evaluated from the optimized Cartesian geometry; the fitted coordinates themselves are the constrained, non-redundant GIC amplitudes.

## 5.4 Cyclopentadiene MSR Benchmark

The cyclopentadiene refinement used Cartesian coordinates and isotopologue rotational data corresponding to the MSR benchmark. The generated coordinate set contained 27 non-redundant symmetry-assigned GICs, of which 10 transformed as $A_1$ and were active. During

Table 7: Glycine II DPCS3-constrained SEfit structural parameters. Bond lengths are in Å; angles and dihedrals are in degrees. Standard errors are propagated from the fitted covariance matrix after projection of the five functional constraints.

| Parameter | Description | Value | Std. error |
|---|---|---|---|
| R(1,2) | C–O(H) | 1.33327 | 0.00273 |
| R(1,3) | C=O | 1.20471 | 0.00297 |
| R(1,5) | C–C | 1.52276 | 0.00223 |
| R(2,4) | O–H | 0.99029 | 0.00246 |
| R(5,6) | C–N | 1.45798 | 0.00354 |
| R(5,7) | C–H7 | 1.08732 | 0.00139 |
| R(5,8) | C–H8 | 1.08742 | 0.00139 |
| R(6,9) | N–H9 | 1.01282 | 0.00540 |
| R(6,10) | N–H10 | 1.01425 | 0.00540 |
| A(2,1,3) | O–C–O | 123.126 | 0.387 |
| A(2,1,5) | O(H)–C–C | 114.228 | 0.130 |
| A(3,1,5) | O=C–C | 122.085 | 0.245 |
| A(1,2,4) | C–O–H | 105.236 | 0.154 |
| A(1,5,6) | C–C–N | 111.940 | 0.193 |
| A(7,5,8) | H–C–H | 106.906 | 0.196 |
| A(9,6,10) | H–N–H | 107.699 | 0.808 |
| D(5,1,2,4) | C–C–O–H | -177.692 | 2.257 |
| D(2,1,5,6) | O(H)–C–C–N | -174.255 | 2.307 |
| D(3,1,5,6) | O=C–C–N | 14.088 | 0.740 |
| D(1,5,6,9) | C–C–N–H9 | 39.622 | 1.819 |
| D(1,5,6,10) | C–C–N–H10 | -84.075 | 1.819 |

the trust-region refinement, accepted steps were checked against the original GIC signature so that the $C_{2v}$ topology and the $A_1$ active space were preserved.

The least-squares problem used moments of inertia and all three principal components for 14 isotopologues. No predicates, parameter classes, or automatic pruning were required. The optimization converged in six accepted steps without rejections, and the optimized-space Hessian was positive definite. The diagnostics are summarized in Table 8.

The largest residual in the rotational-constant representation is 0.229 MHz, while the RMS residual over the 42 corrected rotational constants is 0.0635 MHz. In the fitted moment-of-inertia space, the largest residual is $2.61\times10^{-3}$ amu Å$^2$, and the RMS residual is $7.18\times10^{-4}$ amu Å$^2$. The complete table of corrected experimental rotational constants, calculated constants, and differences is reported in the Supporting Information.

The same 14-isotopologue MSR-like data set was also refined with the symmetry-Cartesian coordinate model. The Cartesian calculation used 27 vibrational displacement directions after removal of translations and rotations, of which 10 were totally symmetric and active. It converged to a minimum with the same numerical rank and essentially the same

Table 8: Cyclopentadiene semiexperimental refinement from the MSR-derived data set. Moments of inertia are in amu Å$^2$ and rotational residuals are in MHz.

| Quantity | Value |
|---|---:|
| Isotopologues | 14 |
| Rotational observables | 42 |
| Final non-redundant GICs | 27 |
| Active $A_1$ GICs | 10 |
| Numerical rank | 10 |
| Accepted/rejected steps | 6/0 |
| Condition number | 454.54 |
| Weighted RMS in inertia space | $7.18 \times 10^{-4}$ |
| Maximum inertia residual | $2.61 \times 10^{-3}$ |
| RMS rotational-constant residual | 0.0635 |
| Maximum rotational-constant residual | 0.2289 |
| Reduced $\chi^2$ | $6.77 \times 10^{-7}$ |
| Smallest Hessian eigenvalue | 0.9788 |
| Stationary point | minimum |

residuals as the GIC refinement, giving a condition number of 635.4 and RMS/maximum rotational-constant residuals of 0.0635/0.2289 MHz. The comparison with the corresponding GIC result is included in Table 11.

Selected structural parameters are reported in Table 9. The uncertainties are propagated from the optimized $A_1$ GIC covariance matrix to topological bond lengths, valence angles, and dihedrals; symmetry-equivalent quantities are condensed.

## 5.5 Cyclic and Polycyclic Systems

The cyclopentadiene benchmark tests a situation where a Z-matrix can be written but cannot treat every ring bond and deformation on the same footing. The automatic GIC model uses endocyclic valence-angle and torsional primitives rather than Cremer–Pople variables or a ring-closing Z-matrix distance. The symmetry-Cartesian run reaches the same rank and residuals without using internal coordinates as active variables. The same principle is then applied to nitrobenzene, azulene, and norcamphor.

Table 9: Selected final topological parameters for cyclopentadiene. Bond lengths are in Å, angles and dihedrals in degrees. Reported uncertainties are one standard deviation.

| Parameter | Atoms | Value |
|---|---|---|
| $R$(C–C) | 1–2 / 1–3 | $1.49921 \pm 0.00029$ |
| $R$(C=C) | 2–4 / 3–5 | $1.34618 \pm 0.00031$ |
| $R$(C–C) | 4–5 | $1.46538 \pm 0.00042$ |
| $R$(C–H) | 1–10 / 1–11 | $1.09446 \pm 0.00015$ |
| $R$(C–H) | 2–6 / 3–7 | $1.07881 \pm 0.00019$ |
| $R$(C–H) | 4–8 / 5–9 | $1.07893 \pm 0.00016$ |
| $A$(C–C–C) | 2–1–3 | $103.069 \pm 0.028$ |
| $A$(C–C–C) | 1–2–4 / 1–3–5 | $109.336 \pm 0.020$ |
| $A$(C–C–C) | 2–4–5 / 3–5–4 | $109.130 \pm 0.014$ |
| $A$(H–C–H) | 10–1–11 | $106.578 \pm 0.021$ |
| $A$(C–C–H) | 1–2–6 / 1–3–7 | $123.942 \pm 0.077$ |
| $A$(C–C–H) | 4–2–6 / 5–3–7 | $126.722 \pm 0.076$ |
| $A$(C–C–H) | 2–4–8 / 3–5–9 | $126.059 \pm 0.044$ |
| $A$(C–C–H) | 5–4–8 / 4–5–9 | $124.811 \pm 0.036$ |
| $D$(C–C–C–C) | 3–1–2–4 | $-179.9928 \pm 0.0000014$ |
| $D$(C–C–C–C) | 1–2–4–5 | $179.9952 \pm 0.0000006$ |
| $D$(C–C–C–C) | 2–4–5–3 | $180.0000 \pm 0.00000001$ |

## 5.6 Nitrobenzene Planar-Observable Benchmark

Nitrobenzene tests two additional features: import of an MSR-style legacy input containing a Z-matrix and dummy atoms, and automatic selection of independent rotational information for a planar molecule. The calculation itself uses the converted Cartesian geometry, not the imported Z-matrix.

The coordinate engine assigned the planar $C_{2v}$ model and generated 36 final non-redundant symmetry-assigned GICs. Thirteen $A_1$ coordinates were active: eight stretching-type and five bending-type coordinates. The planar-component analysis selected the $AB$ pair, i.e., $(I_a, I_b)$ in the moment formulation; the $C$ constants were recomputed from the optimized Cartesian structure and retained as diagnostics.

The fit used nine isotopologues and reports 27 corrected rotational constants, but only the 18 independent $AB$ moment observables entered the normal equations. It converged in four accepted steps without rejections. The final optimized-space Hessian is rank deficient only in the statistical sense expected for this highly correlated planar data set, while all fitted directions have positive curvature. The RMS rotational residual over all reported constants is 0.1480 MHz, with a maximum of 0.2605 MHz for the diagnostic $C$ component of the parent species. Forced $AC$ and $BC$ fits give larger RMS residuals of 0.2593 and 2.4602 MHz, respectively.

Table 10: Nitrobenzene planar semiexperimental benchmark. The fit used the independent $AB$ moment pair; rotational residuals are reported over all diagnostic $A$, $B$, and $C$ constants in MHz.

| Quantity | Value |
|---|---:|
| Isotopologues | 9 |
| Reported corrected constants | 27 |
| Fitted independent components | $AB$ / $(I_a, I_b)$ |
| Final non-redundant GICs | 36 |
| Active $A_1$ GICs | 13 |
| Numerical rank | 13 |
| Accepted/rejected steps | 4/0 |
| Condition number | 27464.44 |
| Weighted RMS in inertia space | $1.32 \times 10^{-2}$ |
| RMS rotational-constant residual | 0.1480 |
| Maximum rotational-constant residual | 0.2605 |
| Reduced $\chi^2$ | $6.26 \times 10^{-4}$ |
| Stationary point | flat or rank deficient |

## 5.7 Azulene Comparison with the MSR Refinement

Azulene is a more demanding polycyclic validation because the MSR analysis uses a step-wise strategy and selected fixed parameters to compensate for incomplete deuterium substitution. The `SEfit` calculation used the same spectroscopic information and structural constraints, but no Z-matrix was built. The MSR-fixed C1–H11/C3–H13 and C2–H12 distances, together with selected fixed valence angles, were supplied as primitive constraints and projected onto the active space.

For the MSR-comparable subset, the diagnostic table contains 12 isotopologues and 36 corrected rotational constants. Because azulene is planar, only the $AB$ pair, corresponding to $(I_a, I_b)$, entered the normal equations; the $C$ constants were retained as diagnostics. The coordinate engine generated 48 non-redundant symmetry-assigned GICs. Seventeen were totally symmetric, and the primitive constraints reduced the effective least-squares space to 10 variables. The topology-validating trust-region solver retained the $C_{2v}$ coordinate model throughout the fit, and the final point was a minimum. The RMS residual over all reported constants was 0.0731 MHz, with a maximum of 0.2900 MHz for the $A$ component of azulene-9-$^{13}$C. Forced $AC$ and $BC$ planar pairs give larger RMS residuals of 0.1086 and 0.3232 MHz.

Cyclopentadiene, azulene, and norcamphor were also used to compare the default GIC coordinate model with the symmetry-Cartesian model. The comparison is summarized in Table 11. In each case, the Cartesian basis spans the same vibrational space as the non-redundant GIC basis after removal of translations and rotations, and the totally symmetric block has the same effective rank after projection of primitive constraints. The rotational residuals are unchanged within numerical precision. The condition number is somewhat system dependent: the GIC model is slightly better conditioned for cyclopentadiene, whereas the symmetry-Cartesian model improves the azulene and norcamphor conditioning. Since the final reported structural parameters are evaluated from the optimized Cartesian geometry and their uncertainties are propagated from the fitted covariance matrix, the lack of direct chemical locality in the working Cartesian amplitudes is not a practical limitation for the final structural output.

Table 11: Comparison of `SEfit` working-coordinate models. "Work." is the number of non-redundant GICs or symmetry-adapted Cartesian vibrational directions, "Sym." is the number of totally symmetric working coordinates before primitive-constraint projection, and "Eff." is the number of independent least-squares variables after projection. Residuals are RMS/maximum absolute values in MHz.

| System | Coordinate model | Work. | Sym. | Eff. | Rank | Cond. | Residuals |
|---|---|---|---|---|---|---|---|
| Cyclopentadiene | GICs | 27 | 10 | 10 | 10 | 454.5 | 0.0635/0.2289 |
| Cyclopentadiene | Sym. Cartesians | 27 | 10 | 10 | 10 | 635.4 | 0.0635/0.2289 |
| Azulene | GICs | 48 | 17 | 10 | 10 | 956.7 | 0.0731/0.2900 |
| Azulene | Sym. Cartesians | 48 | 17 | 10 | 10 | 744.8 | 0.0731/0.2900 |
| Norcamphor | GICs | 48 | 48 | 18 | 18 | 3700.3 | 0.0710/0.2091 |
| Norcamphor | Sym. Cartesians | 48 | 48 | 18 | 18 | 1615.8 | 0.0710/0.2091 |

Selected structural parameters are compared with the reference MSR values in Table 12. The goal is not to reproduce the MSR Z-matrix parameterization, but to verify that the Cartesian/GIC refinement gives the same structural scale without coordinate-path choices. In the original Z-matrix treatment, one ring C–C distance is a ring-closure quantity derived from the remaining coordinates; here all topological bond lengths are evaluated uniformly from the final Cartesian geometry. For the strongly correlated $R(1,9)/R(4,5)$ pair, the MSR values are mapped to the corresponding Cartesian/GIC bonds before differences are formed. The complete parameter comparison and rotational-constant table are in the Supporting Information.

Table 12: Selected azulene structural parameters from the automatic Cartesian/GIC refinement compared with the reference MSR values. Bond lengths are in Å; angles are in degrees. The reported `SEfit` uncertainty is one standard deviation propagated from the optimized active GIC covariance matrix.

| Parameter | `SEfit` | $\sigma$ | MSR | Difference |
|---|---|---|---|---|
| $R(1,9)$ | 1.39745 | 0.00599 | 1.39900 | -0.00155 |
| $R(4,10)$ | 1.37923 | 0.01201 | 1.37800 | +0.00123 |
| $R(4,5)$ | 1.40122 | 0.00653 | 1.40100 | +0.00022 |
| $R(5,6)$ | 1.39054 | 0.00442 | 1.39200 | -0.00146 |
| $R(4,14)$ | 1.08356 | 0.00220 | 1.08700 | -0.00344 |
| $R(5,15)$ | 1.08133 | 0.00230 | 1.08200 | -0.00067 |
| $R(6,16)$ | 1.08117 | 0.00441 | 1.08200 | -0.00083 |
| $A(1,9,10)$ | 106.13001 | 0.52104 | 106.50000 | -0.36999 |
| $A(5,6,7)$ | 130.09492 | 0.17695 | 129.90000 | +0.19492 |
| $A(6,5,15)$ | 115.90333 | 0.30686 | 115.70000 | +0.20333 |
| $A(4,10,9)$ | 127.46958 | 0.08847 | 127.58000 | -0.11042 |
| $A(5,6,16)$ | 114.95254 | 0.08847 | 115.07000 | -0.11746 |

## 5.8 Norcamphor Bridged-Topology Stress Test

Norcamphor is a non-planar bridged-molecule stress test combining $C_1$ topology, a carbonyl group, fused ring constraints, and no deuterium-substituted isotopologues. The input geometry was a computed Cartesian structure derived from the published Supporting Information geometry. The isotopologue data consisted of the parent species, all seven singly substituted

$^{13}$C species, and the $^{18}$O isotopologue. The ground-state rotational constants were corrected with computed vibrational contributions using the additive convention $B_e = B_0 + \Delta B_{\mathrm{vib}}$.

Because norcamphor is $C_1$, symmetry does not reduce the active space: all GICs are totally symmetric. The coordinate engine generated 48 final non-redundant GICs, comprising 18 stretches, 25 bends, and 5 torsional/ring coordinates. The reduced-dimensionality MSR model fixes hydrogen positions through Z-matrix variables; here the same information was translated into 30 primitive constraints, one C–H stretch, one C–C–H angle, and one C–C–C–H dihedral per hydrogen. Projection of these constraints reduced the least-squares space from 48 to 18 variables, corresponding to the heavy-atom skeleton.

The equal-weight moment-of-inertia refinement converged in eight accepted steps without rejections, and the final point was a minimum. The RMS residual over the 27 corrected rotational constants was 0.0710 MHz, with a maximum of 0.2091 MHz. The published reduced-dimensionality MSR refinement reports a standard deviation of 0.05 MHz for the reduced-dimensionality reference model. Here the residuals are dominated by the $A$ constants of the $^{13}$C1 and $^{13}$C2 isotopologues, while the residual mean for each rotational component remains close to zero. Because the present benchmark uses the printed vibrational corrections, rounded to 0.1 MHz, the remaining difference should not be overinterpreted. A fully homogeneous comparison would require unrounded corrections and the exact statistical convention of the original MSR run. An automatic local-frame hydrogen constraint variant gives a slightly smaller RMS residual of 0.0673 MHz, but the tabulated comparison uses the Supporting Information (SI) Z-matrix-equivalent primitive constraints.

As an independent geometrical diagnostic, Kraitchman substitution coordinates were computed for the singly substituted heavy-atom isotopologues and compared with the absolute coordinates of the fitted structure in the parent principal-axis frame. This comparison was not used in the least-squares objective. It provides a direct check that the fitted Cartesian geometry is consistent with the substitution information. The RMS absolute-coordinate difference over the 24 axis components is 0.02097 Å, and the largest component difference is

0.06231 Å for the $a$ coordinate of the $^{13}$C1 isotopologue (Table 13).

Table 13: Kraitchman diagnostic for the norcamphor refinement. RMS and maximum deviations compare absolute Kraitchman substitution coordinates with absolute fitted coordinates over the $a$, $b$, and $c$ principal axes. Values are in Å.

| Isotopologue | Atom | RMS | Max. | Axis |
|---|---|---|---|---|
| $^{13}$C1 | C1 | 0.03716 | 0.06231 | $a$ |
| $^{13}$C2 | C2 | 0.02043 | 0.02777 | $b$ |
| $^{13}$C3 | C3 | 0.01640 | 0.02080 | $c$ |
| $^{13}$C4 | C4 | 0.02469 | 0.04217 | $c$ |
| $^{13}$C5 | C5 | 0.02139 | 0.03468 | $b$ |
| $^{13}$C6 | C6 | 0.01452 | 0.02268 | $c$ |
| $^{13}$C7 | C7 | 0.00638 | 0.00769 | $a$ |
| $^{18}$O | O8 | 0.01154 | 0.01591 | $c$ |

The symmetry-Cartesian refinement was repeated with the same 30 primitive hydrogen constraints. Because norcamphor is $C_1$, all 48 vibrational directions are totally symmetric before constraint projection; the constraints again reduce the fit to 18 effective variables. The Cartesian calculation converges in eight accepted steps to a minimum, gives the same rank and RMS/maximum rotational residuals of 0.0710/0.2091 MHz, and lowers the condition number from 3700.3 to 1615.8.

## 5.9 Refinements with Parameter Classes

Parameter classes provide a practical strategy for semiexperimental refinements in which the available isotopic information is insufficient to determine all structural variables independently. Hydrogen-containing distances or angles can be refined collectively or held fixed as symmetry-complete primitive-coordinate relations. The black-box character of the workflow therefore does not eliminate chemical judgment; it confines that judgment to constraints, predicates, and classes that have an unambiguous internal-coordinate meaning.

Table 14: Norcamphor $C_1$ reduced-dimensionality stress test. Moments of inertia were used as fit observables; rotational residuals are reported in MHz.

| Quantity | Value |
|---|---|
| Isotopologues | 9 |
| Rotational observables | 27 |
| Final non-redundant GICs | 48 |
| Totally symmetric GICs | 48 |
| Primitive hydrogen constraints | 30 |
| Effective independent variables | 18 |
| Numerical rank | 18 |
| Accepted/rejected steps | 8/0 |
| Condition number | 3700.28 |
| Weighted RMS in inertia space | $4.94 \times 10^{-3}$ |
| RMS rotational-constant residual | 0.0710 |
| Maximum rotational-constant residual | 0.2091 |
| Reduced $\chi^2$ | $7.31 \times 10^{-5}$ |
| Stationary point | minimum |

# 6 Conclusions

`SEfit` separates the experimental refinement model from the coordinate construction problem. Molecular structures enter and leave the workflow as Cartesian geometries and primitive internal parameters, while the least-squares solver operates in a non-redundant totally symmetric active space. The active space can be built either from symmetry-adapted GICs or from translation- and rotation-free symmetry Cartesians; constraints, predicates, and parameter classes are specified in the same primitive-coordinate layer in both cases.

The benchmarks show that this separation is especially useful when a compact acyclic Z-matrix is no longer natural: constrained glycine I/II, cyclic cyclopentadiene and nitrobenzene, fused azulene, and bridged norcamphor all retain well-defined ranks, symmetry, and propagated structural uncertainties. Planar molecules are handled with two independent rotational components and the third as a diagnostic. When both active spaces are applicable, they give the same effective dimensions and residuals.

Thus the essential requirements for black-box semiexperimental refinement are a complete non-redundant symmetry-consistent active space, internal-coordinate constraints that can be projected onto that space, and rigorous covariance propagation from fitted amplitudes to reported structural parameters. The implementation provides these ingredients while remaining compatible with standard correction workflows, automated starting geometries, and reduced-dimensionality treatments of incomplete isotopic data.

## Supporting Information Available

Complete glycine I, glycine II, cyclopentadiene, nitrobenzene, and azulene rotational-constant comparisons, including corrected experimental constants, calculated constants, and differences. The nitrobenzene Kraitchman diagnostic is also provided. Additional diagnostic tables for glycolaldehyde and norcamphor, including final Cartesian coordinates, propagated topological parameters, and residuals, are generated by the archived `SEfit` output.

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